

Fourier Series

①

Bernoulli's rule of integration by parts

$$\int uv \, dx = u \int v - u' \iint v + u'' \iiint v - \dots$$

where $u', u'', u''' \dots$ are successive derivatives of u .

Standard Results

- 1) $\sin(-\theta) = -\sin\theta$, $\cos(-\theta) = \cos\theta$.
- 2) If 'n' is an integer, then
 - i) $\sin(n\pi) = 0$
 - ii) $\cos(n\pi) = (-1)^n = \begin{cases} +1 & , n \text{ is even} \\ -1 & , n \text{ is odd} \end{cases}$
- 3) $\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$
 $\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$
- 4) $\int_{-a}^a f(x) \, dx = \begin{cases} 2 \int_0^a f(x) \, dx & \text{if } f(x) \text{ is even} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases}$
 $\int_0^{2a} f(x) \, dx = \begin{cases} 2 \int_0^a f(x) \, dx & \text{if } f(x) \text{ is even} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases}$

- 5) Even function $\times$ Even function = Even function
Odd function $\times$ Odd function = Even function
Even function $\times$ Odd function = Odd function

Periodic function

A function $f(x)$ is said to be periodic function with period T , if $f(x+T) = f(x) \quad \forall x, T > 0$.

Ex $f(x) = \sin x$ is periodic function with period 2π

Fourier series

Fourier series

The Fourier series of a periodic function with period $2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

where a_0, a_n, b_n are the Fourier coefficients given by.

Interval	a_0	a_n	b_n
$(-l, l)$	$\frac{1}{2l} \int_{-l}^l f(x) dx$	$\frac{1}{2l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$	$\frac{1}{2l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$
$(-\pi, \pi)$	$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$	$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$	$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$
$(0, 2l)$	$\frac{1}{l} \int_0^{2l} f(x) dx$	$\frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$	$\frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$
$(0, 2\pi)$	$\frac{1}{\pi} \int_0^{2\pi} f(x) dx$	$\frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx$	$\frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx$

Note:- 1) If $f(x)$ is discontinuous function of the form

$$i) f(x) = \begin{cases} \phi(x) & \text{in } -l < x < 0 \\ \psi(x) & \text{in } 0 < x < l \end{cases}, \text{ then}$$

$f(x)$ is even if $\phi(-x) = \psi(x)$ e

$f(x)$ is odd if $\phi(-x) = -\psi(x)$

$$ii) f(x) = \begin{cases} \phi(x) & \text{in } 0 < x < l \\ \psi(x) & \text{in } l < x < 2l \end{cases}, \text{ then}$$

$f(x)$ is even if $\phi(2l-x) = \psi(x)$ e

$f(x)$ is odd if $\phi(2l-x) = -\psi(x)$

2) If $f(x)$ is discontinuous at x , then at x

the fourier series converges to $\frac{1}{2} [f(x^+) + f(x^-)]$

where $f(x^+)$ & $f(x^-)$ are Right hand & left hand limits of $f(x)$ respectively.

Fourier Co-efficients for Even (or) odd function

Interval	Nature	a_0	a_n	b_n
①	<u>Even function</u>			
$(-l, l)$	$f(-x) = f(x)$	$\frac{2}{l} \int_0^l f(x) dx$	$\frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$	0
$(0, 2l)$	$f(2l-x) = f(x)$			
$(-\pi, \pi)$	$f(-x) = f(x)$	$\frac{2}{\pi} \int_0^{\pi} f(x) dx$	$\frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$	0
$(0, 2\pi)$	$f(2\pi-x) = f(x)$			
②	<u>Odd function</u>			
$(-l, l)$	$f(-x) = -f(x)$	0	0	$\frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$
$(0, 2l)$	$f(2l-x) = -f(x)$			
$(-\pi, \pi)$	$f(-x) = -f(x)$	0	0	$\frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx$
$(0, 2\pi)$	$f(2\pi-x) = -f(x)$			

Problems

1) Find the fourier series of $f(x) = |x|$ in $-\pi < x < \pi$, Hence deduce that $\frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$

Soln:- Given function is $f(x) = |x|$

To check function is even (or) odd.

we have $f(x) = |x|$ in $(-\pi, \pi)$

$$f(-x) = |-x| = |x| = f(x).$$

$$\text{ie } f(-x) = f(x).$$

$\therefore f(x)$ is even function

$$\Rightarrow \boxed{b_n = 0}$$

$$\text{Here } f(x) = |x| = \begin{cases} -x & \text{in } -\pi < x < 0 \\ +x & \text{in } 0 < x < \pi \end{cases}$$

$$\therefore a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \quad \text{ie } f(x) = |x| = x \text{ in } 0 < x < \pi$$

$$= \frac{2}{\pi} \int_0^{\pi} x dx$$

$$= \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi}$$

$$= \frac{2}{\pi(2)} \times (x^2)_0^{\pi}$$

$$= \frac{1}{\pi} (\pi^2 - 0) = \frac{\pi^2}{\pi} = \pi$$

$$\therefore \boxed{a_0 = \pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos(nx) \, dx$$

$$= \frac{2}{\pi} \left[\left(x \frac{\sin(nx)}{n} \right) - (1) \left(-\frac{\cos nx}{n^2} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{1}{n^2} (\cos nx)_0^{\pi} \right]$$

$$= \frac{2}{n^2 \pi} (\cos nx)_0^{\pi}$$

$$= \frac{2}{n^2 \pi} (\cos n\pi - \cos 0)$$

$$\therefore a_n = \frac{2}{n^2 \pi} ((-1)^n - 1)$$

$$\Rightarrow \boxed{a_n = -\frac{2}{n^2 \pi} [1 - (-1)^n]}$$

$$\text{or } \boxed{b_n = 0}$$

Fourier series of $f(x)$ with period 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{-2}{n^2 \pi} [1 - (-1)^n] \cos nx \quad \text{--- (1)}$$

is the required Fourier series for given $f(x)$.

Deduction

(4)

Put $x=0$ in eq (1).

$$f(0) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{-2}{n^2\pi} [1 - (-1)^n] \times 1$$

$$0 = \frac{\pi}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n]$$

$$-\frac{\pi}{2} = -\frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} (1 - (-1)^n)$$

$$\frac{\pi^2}{4} = \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n]$$

$$(\because [1 - (-1)^n] = \begin{cases} 2 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases})$$

$$\frac{\pi^2}{4} = \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} (2)$$

$$\frac{\pi^2}{8} = \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$\Rightarrow \boxed{\frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}}$$

2) obtain the fourier series for $f(x) = \frac{\pi-x}{2}$ in $0 < x < 2\pi$. Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ (5)

Solun: Given $f(x) = \frac{\pi-x}{2}$ is defined in $(0, 2\pi)$

To check $f(x)$ is even or odd function

$$\begin{aligned} f(2\pi-x) &= \frac{\pi-(2\pi-x)}{2} = \frac{\pi-2\pi+x}{2} = \frac{-\pi+x}{2} \\ &= -\left(\frac{\pi-x}{2}\right) \\ &= -f(x) \end{aligned}$$

$$\text{ie } f(-x) = -f(x)$$

$\therefore$ Given function $f(x)$ is odd function
clearly $\boxed{a_0=0}$ & $\boxed{a_n=0}$

$$\begin{aligned} \text{Now } b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi-x}{2}\right) \sin(nx) dx \\ &= \frac{2}{\pi} \cdot \frac{1}{2} \int_0^{\pi} (\pi-x) \sin(nx) dx \\ &= \frac{1}{\pi} \left[(\pi-x) \left(\frac{-\cos nx}{n} \right) - (1) \left(\frac{-\sin nx}{n^2} \right) \right]_0^{\pi} \\ &= \frac{1}{\pi} \left(-\frac{1}{n} (\pi-x) \cos nx \right)_0^{\pi} \\ &= \frac{-1}{n\pi} \left((\pi-\pi) \cos n\pi - (\pi-0) \cos(0) \right) \end{aligned}$$

$$= -\frac{1}{n\pi} (0 - \pi)$$

$$= -\frac{1}{n\pi} (-\pi) = \frac{1}{n}$$

$$\Rightarrow \boxed{b_n = \frac{1}{n}}$$

Thus fourier series we have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= 0 + 0 + \sum_{n=1}^{\infty} \frac{1}{n} \sin nx$$

$$\frac{\pi - x}{2} = \sum_{n=1}^{\infty} \frac{1}{n} \sin(nx) \quad \text{--- ①}$$

∴ the required fourier series of $f(x)$

put $x = \pi/2$ in eq ①.

$$\text{①} \Rightarrow \frac{\pi - \pi/2}{2} = \sum_{n=1}^{\infty} \frac{1}{n} \sin(n\pi/2)$$

$$\frac{2\pi - \pi}{4} = \frac{\sin \pi/2}{1} + \frac{\sin \pi}{2} + \frac{\sin(3\pi/2)}{3} + \frac{\sin 2\pi}{4} + \frac{\sin(5\pi/2)}{5} + \dots$$

$$\boxed{\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \dots}$$

3) Find the fourier series of $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in

$0 < x < 2\pi$, Hence. Evaluate $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

Soln:- Given function is $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in $(0, 2\pi)$

To check $f(x)$ is even or odd

$$f(x) = \left(\frac{\pi-x}{2}\right)^2 \text{ in } (0, 2\pi)$$

$$\begin{aligned} \text{Consider } f(2\pi-x) &= \left(\frac{\pi - (2\pi-x)}{2}\right)^2 = \left(\frac{\pi - 2\pi + x}{2}\right)^2 \\ &= \left(\frac{-\pi + x}{2}\right)^2 = \left[-\left(\frac{\pi-x}{2}\right)\right]^2 \\ &= \left(\frac{\pi-x}{2}\right)^2 = f(x) \end{aligned}$$

$$\text{ie } f(2\pi-x) = f(x)$$

$\therefore$ Given ^{function} $f(x)$ is even function.

$$\Rightarrow \boxed{b_n = 0}$$

$$\text{Now } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi-x}{2}\right)^2 dx$$

$$= \frac{2}{4\pi} \int_0^{\pi} (\pi-x)^2 dx$$

$$= \frac{1}{2\pi} \left[\frac{(\pi-x)^3}{-3} \right]_0^{\pi}$$

$$= -\frac{1}{6\pi} \left[(\pi-x)^3 \right]_0^{\pi} = -\frac{1}{6\pi} \left((\pi-\pi)^3 - (\pi-0)^3 \right)$$

$$= -\frac{1}{6\pi} (0^3 - \pi^3) = \frac{\pi^3}{6\pi} = \frac{\pi^2}{6}$$

$$\Rightarrow \boxed{a_0 = \frac{\pi^2}{6}}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi-x}{2}\right)^2 \cos(nx) dx$$

$$= \frac{2}{4\pi} \int_0^{\pi} (\pi-x)^2 \cos(nx) dx$$

$$= \frac{1}{2\pi} \left[(\pi-x)^2 \left(\frac{\sin(nx)}{n} \right) - (-2)(\pi-x) \left(-\frac{\cos(nx)}{n^2} \right) + 2 \left(-\frac{\sin(nx)}{n^3} \right) \right]_0^{\pi}$$

$$= -\frac{1}{2\pi} \left[2(\pi-x) \frac{\cos(nx)}{n^2} \right]_0^{\pi}$$

$$= -\frac{2}{2n^2\pi} \left[(\pi-x) \cos(nx) \right]_0^{\pi}$$

$$= -\frac{1}{n^2\pi} \left((\pi-\pi) \cos n\pi - \pi \cos 0 \right)$$

$$= -\frac{1}{n^2\pi} (0 - \pi) = \frac{\pi}{n^2\pi} = \frac{1}{n^2}$$

$$\boxed{a_n = \frac{1}{n^2}}$$

The fourier series of $f(x)$ with period 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx).$$

$$f(x) = \frac{\pi^2/6}{2} + \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx$$

$$f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx \quad \text{--- ①.}$$

is the required fourier series.

Deduction

put $x=0$ in eq ①

$$f(0) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{1}{n^2} \times 1.$$

$$\left(\frac{\pi-0}{2}\right)^2 = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{4} - \frac{\pi^2}{12} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

||

4) Find the fourier series of

$$f(x) = \begin{cases} 1 + \frac{2x}{\pi} & \text{in } -\pi < x \leq 0 \\ 1 - \frac{2x}{\pi} & \text{in } 0 \leq x < \pi \end{cases}$$

Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Solun:- Given $f(x)$ is defined in $(-\pi, \pi)$

To check $f(x)$ is even or odd function

$$\phi(x) = 1 + \frac{2x}{\pi}, \quad \psi(x) = 1 - \frac{2x}{\pi}$$

$$\text{Consider } \phi(-x) = 1 + \frac{2(-x)}{\pi} = 1 - \frac{2x}{\pi} = \psi(x)$$

$$\text{ie } \phi(-x) = \underline{\underline{\psi(x)}}$$

$\therefore f(x)$ is even function

$$\text{clearly } \boxed{b_n = 0}$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi}\right) dx$$

$$= \frac{2}{\pi} \left(x - \frac{2x^2}{2\pi} \right)_0^{\pi}$$

$$= \frac{2}{\pi} \left(x - \frac{x^2}{\pi} \right)_0^{\pi} = \frac{2}{\pi} \left(\pi - \frac{\pi^2}{\pi} \right) = \frac{2}{\pi} (\pi - \pi)$$

$$\Rightarrow \boxed{a_0 = 0}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi}\right) \cos nx dx$$

$$= \frac{2}{\pi} \left[\left(1 - \frac{2x}{\pi}\right) \left(\frac{\sin nx}{n}\right) - \left(-\frac{2}{\pi}\right) \left(\frac{-\cos nx}{n^2}\right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left(-\frac{2}{\pi n^2} (\cos nx) \right)_0^{\pi}$$

$$= \frac{-4}{\pi^2 n^2} (\cos nx)_0^{\pi}$$

$$= \frac{-4}{\pi^2 n^2} (\cos n\pi - \cos 0)$$

$$= \frac{-4}{\pi^2 n^2} ((-1)^n - 1)$$

$$\boxed{a_n = \frac{4}{\pi^2 n^2} (1 - (-1)^n)}$$

We have fourier series of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{4}{\pi^2 n^2} (1 - (-1)^n) \cos nx$$

$$= \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} (1 + (-1)^n) \cos(nx) \quad \text{--- (1)} \quad (7)$$

is the required fourier series

Put $x=0$ in Eq (1)

$$\therefore 1 = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2}$$

$$1 = \frac{4}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{2}{n^2}$$

$$1 = \frac{8}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{8} = \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Hence the deduction.

5) Obtain the fourier series for the function

$$f(x) = \begin{cases} \pi x & \text{in } 0 \leq x \leq 1 \\ \pi(2-x) & \text{in } 1 \leq x \leq 2 \end{cases}$$

Deduce that $\frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$

Solun:- Here $f(x)$ is defined in $[0, 2]$
Comparing this with $[0, 2l]$

$$\Rightarrow 2l = 2$$
$$\Rightarrow \boxed{l=1}$$

$$\phi(x) = \pi x ; \quad \psi(x) = \pi(2-x)$$

To check $f(x)$ is even or odd function.

$$\phi(2-x) = \pi(2-x)$$
$$= \psi(x)$$

$\therefore f(x)$ is even function

clearly $\boxed{b_n = 0}$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$= \frac{2}{1} \int_0^1 f(x) dx = 2 \int_0^1 \pi x dx$$

$$= 2 \left[\frac{\pi x^2}{2} \right]_0^1$$

$$= 2 \frac{\pi}{2} (x^2)_0^1$$

$$= \pi(1)$$

$$\boxed{a_0 = \pi}$$

(8)

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{1} \int_0^1 f(x) \cos(n\pi x) dx$$

$$= 2 \int_0^1 x \cos(n\pi x) dx$$

$$= 2\pi \int_0^1 x \cos(n\pi x) dx$$

$$= 2\pi \left[x \frac{\sin(n\pi x)}{(n\pi)} - (1) \left(-\frac{\cos(n\pi x)}{(n\pi)^2} \right) \right]_0^1$$

$$= 2\pi \left(\frac{1}{n^2\pi^2} (\cos n\pi x)' \right)_0^1$$

$$= \frac{2\pi}{n^2\pi^2} (\cos n\pi - \cos 0)$$

$$= \frac{2}{n^2\pi} ((-1)^n - 1)$$

$$a_n = \frac{-2}{n^2\pi} (1 - (-1)^n)$$

We have fourier series of $f(x)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{-2}{n^2\pi} (1 - (-1)^n) \cos(n\pi x)$$

$$= \frac{\pi}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2} \cos(n\pi x)$$

$$f(x) = \frac{\pi}{2} - \frac{2}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{2}{n^2} \cos(n\pi x) \quad \left(\because 1-(-1)^n = \begin{cases} 0 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd} \end{cases} \right)$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos(n\pi x) \quad \text{--- ①}$$

is the required fourier series

Put $x=0$ in Eq ①

$$\therefore ① \Rightarrow 0 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$-\frac{\pi}{2} = -\frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{8} = \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$\Rightarrow \frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$$

b) obtain the fourier series of $f(x) = e^{-ax}$ in the interval $(-\pi, \pi)$. Deduce that $\frac{\pi}{\sinh \pi} = 2 \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2+1}$

Soln:- Given function is
 $f(x) = e^{-ax}$

To check the function is even (or) odd

$$f(x) = e^{-ax} \quad \text{in } (-\pi, \pi)$$

$$f(-x) = e^{ax} \neq f(x) \\ \neq -f(x)$$

$\therefore f(x)$ is neither even nor odd

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} dx \\ &= \frac{1}{\pi} \left(\frac{e^{-ax}}{-a} \right)_{-\pi}^{\pi} \\ &= -\frac{1}{a\pi} \left(e^{-a\pi} - e^{a\pi} \right) \\ &= -\frac{1}{a\pi} (e^{-a\pi} - e^{a\pi}) \\ &= \frac{2}{a\pi} \left[\frac{e^{a\pi} - e^{-a\pi}}{2} \right] \\ &= \frac{2}{a\pi} (\sinh a\pi) \end{aligned}$$

$$\therefore \boxed{a_0 = \frac{2 \sinh(a\pi)}{a\pi}}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{a^2 + n^2} (-a \cos nx + n \sin nx) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi(a^2 + n^2)} [a e^{-ax} \cos nx]_{-\pi}^{\pi}$$

$$= \frac{a}{\pi(a^2 + n^2)} (-e^{-a\pi} \cos n\pi + e^{a\pi} \cos n\pi)$$

$$= \frac{a \cos n\pi}{\pi(a^2 + n^2)} (-e^{-a\pi} + e^{a\pi})$$

$$= \frac{2a \cos n\pi}{\pi(a^2 + n^2)} \left(\frac{e^{a\pi} - e^{-a\pi}}{2} \right)$$

$$\boxed{a_n = \frac{2a \cos n\pi}{\pi(a^2 + n^2)} (\sinh(a\pi))}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \sin(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{a^2 + n^2} (-a \sin nx - n \cos nx) \right]_{-\pi}^{\pi}$$

$$= \frac{-n}{\pi(a^2 + n^2)} [e^{-ax} \cos nx]_{-\pi}^{\pi}$$

$$= \frac{-\eta}{\pi(a^2+n^2)} \left(e^{a\pi} \cos(n\pi) - e^{-a\pi} \cos(n\pi) \right)$$

$$= \frac{-\eta \cos(n\pi)}{\pi(a^2+n^2)} \left(e^{-a\pi} - e^{a\pi} \right)$$

$$= \frac{\eta (-1)^n \times 2}{\pi(a^2+n^2)} \left(\frac{e^{a\pi} - e^{-a\pi}}{2} \right)$$

$$\boxed{b_n = \frac{2\eta(-1)^n \sinh(a\pi)}{\pi(a^2+n^2)}}$$

Fourier series of $f(x)$ with period 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{\sinh(a\pi)}{a\pi} + \sum_{n=1}^{\infty} \frac{2a(-1)^n}{\pi(a^2+n^2)} \sinh a\pi \cdot \cos nx + \sum_{n=1}^{\infty} \frac{2\eta(-1)^n}{\pi(a^2+n^2)} \sinh a\pi \cdot \sin nx$$

is the required fourier series

— ①

Deduction

put $x=0$ $a=1$

$$f(0) = \frac{\sinh(\pi)}{\pi} + \sum_{n=1}^{\infty} \frac{2(-1)^n}{\pi(1^2+n^2)} \sinh \pi \times 1$$

$$e^0 = \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(n^2+1)}$$

$$1 = \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \left[-\frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{(n^2+1)} \right]$$

$$= \frac{\sinh \pi}{\pi} - \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(n^2+1)}$$

$$1 = \frac{2 \sinh \pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(n^2+1)}$$

$$\boxed{\frac{\pi}{\sinh \pi} = 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{(n^2+1)}}$$

7) Find the fourier series in $(-\pi, \pi)$ to represent $f(x) = x - x^2$. Hence deduce $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$

Soln:- Given $f(x) = x - x^2$ in $(-\pi, \pi)$

To check the function is even (or) odd

$$f(x) = x - x^2 \text{ in } (-\pi, \pi)$$

$$f(-x) = -x - (-x)^2 = -x - x^2 \neq f(x)$$

$$\neq -f(x)$$

$\therefore$ Given function $f(x)$ is neither even nor odd

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x - x^2) dx = \frac{1}{\pi} \left(\frac{x^2}{2} - \frac{x^3}{3} \right)_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{1}{2} (x^2)_{-\pi}^{\pi} - \frac{1}{3} (x^3)_{-\pi}^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{2} (\pi^2 - \pi^2) - \frac{1}{3} (\pi^3 - (-\pi)^3) \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{2}(0) - \frac{1}{3} (\pi^3 + \pi^3) \right]$$

$$= \frac{1}{\pi} \left(-\frac{2}{3} \pi^3 \right) = -\frac{2}{3} \pi^2 \Rightarrow \boxed{a_0 = -\frac{2\pi^2}{3}}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x-x^2) \cos(nx) dx$$

$$= \frac{1}{\pi} \left[(x-x^2) \left(\frac{\sin(nx)}{n} \right) - (1-2x) \left(-\frac{\cos(nx)}{n^2} \right) + (-2) \left(\frac{\sin(nx)}{n^3} \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n^2} [(1-2x) \cos(nx)] \right\}_{-\pi}^{\pi}$$

$$= \frac{1}{\pi n^2} [(1-2\pi) \cos n\pi] - [(1+2\pi) \cos n\pi]$$

$$= \frac{\cos n\pi}{n^2 \pi} (1-2\pi - 1-2\pi)$$

$$= \frac{(-1)^n}{n^2 \pi} (-4\pi) \Rightarrow \boxed{a_n = \frac{-4(-1)^n}{n^2}}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x-x^2) \sin(nx) dx$$

$$= \frac{1}{\pi} \left[(x-x^2) \left(-\frac{\cos(nx)}{n} \right) - (1-2x) \left(\frac{\sin(nx)}{n^2} \right) + (-2) \left(\frac{\cos(nx)}{n^3} \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\left(-\frac{1}{n} \right) (x-x^2) (\cos(nx)) - \frac{2}{n^3} \cos(nx) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{1}{n} \{ (\pi-\pi^2) \cos n\pi \} - (-\pi-\pi^2) \cos(-n\pi) \right] - \frac{2}{n^3} \{ \cos(n\pi) - \cos(-n\pi) \}$$

$$= \frac{1}{\pi} \left[-\frac{\cos n\pi}{n} (\pi - \pi^2 + \pi + \pi^2) \right]$$

$$= \frac{-(-1)^n}{n\pi} (2\pi) \Rightarrow \boxed{b_n = \frac{-2(-1)^n}{n}}$$

∴ Required fourier series for given $f(x)$ with period 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$= \frac{-2\pi^2/3}{2} + \sum_{n=1}^{\infty} \frac{-4(-1)^n}{n^2} \cos(nx) + \sum_{n=1}^{\infty} \frac{-2(-1)^n}{n} \sin(nx)$$

$$= -\frac{\pi^2}{3} - \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos(nx) - \sum_{n=1}^{\infty} \frac{2(-1)^n}{n} \sin(nx) \quad \text{--- (1)}$$

Deduction :- put $x=0$ in eq (1)

$$f(0) = -\frac{\pi^2}{3} - \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \times 1$$

$$0 = -\frac{\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$\frac{\pi^2}{3} = -4 \left[\frac{-1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots \right]$$

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

8) Find the Fourier series of the function

$$f(x) = \begin{cases} x & \text{in } 0 \leq x \leq \pi \\ 2\pi - x & \text{in } \pi \leq x \leq 2\pi \end{cases} \quad \text{Hence deduce}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Soln 1- To check Given function is even (or) odd

$$\text{Let } \phi(x) = x \quad ; \quad \psi(x) = 2\pi - x \quad \text{in } (0, 2\pi)$$

$$\text{Consider } \phi(2\pi - x) = 2\pi - x = \psi(x)$$

$\therefore f(x)$ is an even function

$$\therefore \boxed{b_n = 0}$$

$$a_0 = \frac{2}{\pi} \int_0^\pi f(x) dx = \frac{2}{\pi} \int_0^\pi x dx = \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^\pi = \frac{2}{\pi} \left(\frac{\pi^2}{2} - 0 \right)$$

$$\boxed{a_0 = \pi}$$

$$a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^\pi x \cos(nx) dx$$

$$= \frac{2}{\pi} \left[x \left(\frac{\sin(nx)}{n} \right) - \left(\frac{\cos(nx)}{n^2} \right) \right]_0^\pi$$

$$= \frac{2}{\pi} \left[\frac{1}{n^2} (\cos(nx)) \right]_0^\pi$$

$$= \frac{2}{n^2 \pi} (\cos n\pi - \cos 0) = \frac{2}{n^2 \pi} ((-1)^n - 1)$$

$$\boxed{a_n = \frac{-2}{n^2 \pi} [1 - (-1)^n]}$$

The fourier series of $f(x)$ with period 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{-2}{\pi n^2} (1 - (-1)^n) \cos(nx) \quad \text{--- ①}$$

ix required fourier series

Put $x=0$

$$f(0) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{-2}{\pi n^2} (1 - (-1)^n)$$

$$0 = \frac{\pi}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(1 - (-1)^n)}{n^2} \quad \left(\because f(0) = 0 \quad \text{b.c.z} \right. \\ \left. f(x) = x \quad \text{in } (0, \pi) \right)$$

$$\frac{\pi}{2} = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2}$$

$$\frac{\pi^2}{4} = \sum_{n=1,3,5,\dots}^{\infty} \frac{2}{n^2}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Q) obtain the fourier series for the function

$$f(x) = \begin{cases} -\pi & \text{in } -\pi < x < 0 \\ x & \text{in } 0 < x < \pi \end{cases}$$

Hence deduce that, sum of the reciprocal squares of the odd integers is equal to $\frac{\pi^2}{8}$.

Solution To check function is even (or) odd function

$$\text{Let } \phi(x) = -\pi, \quad \psi(x) = x \quad \text{in } (-\pi, \pi)$$

$$\begin{aligned} \phi(-x) &= -\pi \neq \psi(x) \\ &\neq -\psi(x) \end{aligned}$$

$\therefore f(x)$ is neither even nor odd function.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) dx + \int_0^{\pi} f(x) dx \right]$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 (-\pi) dx + \int_0^{\pi} x dx \right]$$

$$= \frac{1}{\pi} \left[-\pi(x)_{-\pi}^0 + \left(\frac{x^2}{2} \right)_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[-\pi(+\pi) + \frac{\pi^2}{2} \right] = \frac{1}{\pi} \left(-\pi^2 + \frac{\pi^2}{2} \right)$$

$$= -\frac{1}{\pi} \frac{\pi^2}{2}$$

$$\boxed{a_0 = -\pi/2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \cos(nx) dx + \int_0^{\pi} f(x) \cos(nx) dx \right]$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \cos(nx) dx + \int_0^{\pi} x \cos(nx) dx \right]$$

$$= \frac{1}{\pi} \left[-\pi \left\{ \frac{\sin(nx)}{n} \right\}_{-\pi}^0 + \left\{ x \frac{\sin(nx)}{n} - 1 \times \left(-\frac{\cos(nx)}{n} \right) \right\}_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n^2} [\cos(nx)]_0^{\pi} \right\}$$

$$= \frac{1}{n^2 \pi} (\cos n\pi - \cos 0)$$

$$= \frac{1}{n^2 \pi} \{ (-1)^n - 1 \} \Rightarrow \boxed{a_n = -\frac{1}{n^2 \pi} \{ 1 - (-1)^n \}}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \sin(nx) dx + \int_0^{\pi} f(x) \sin(nx) dx \right]$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \sin(nx) dx + \int_0^{\pi} x \sin(nx) dx \right]$$

$$= \frac{1}{\pi} \left[-\pi \left(-\frac{\cos(nx)}{n} \right)_{-\pi}^0 + \left(x \left(-\frac{\cos(nx)}{n} \right) - 1 \left(-\frac{\sin nx}{n^2} \right) \right)_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi}{n} (-\cos n\pi) - \frac{1}{n} (\pi \cos n\pi - 0) \right]$$

$$= \frac{1}{\pi} \times \frac{\pi}{n} (1 - \cos n\pi - \cos n\pi)$$

$$= \frac{1}{n} [1 - 2\cos n\pi]$$

$$\boxed{b_n = \frac{1}{n} [1 - 2(-1)^n]}$$

the fourier series of function $f(x)$ in the interval 2π is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{-\pi}{4} + \sum_{n=1}^{\infty} \frac{-1}{n^2\pi} [1 - (-1)^n] \cos(nx) + \sum_{n=1}^{\infty} \frac{1}{n} [1 - 2(-1)^n] \sin nx$$

— ①

is the required fourier series

Deduction put $x=0$ in eq ①

It is clear that $f(x)$ at $x=0$ is a point of discontinuity.

Hence series converges to $\frac{1}{2} [f(0^+) + f(0^-)]$

Since $f(0^+) = 0 \quad \therefore f(x) = x$ in $(0, \pi)$

$f(0^-) = -\pi \quad \therefore f(x) = -\pi$ in $(-\pi, 0)$

$$\therefore f(0) = \frac{1}{2} [f(0^+) + f(0^-)] = \frac{1}{2} (0 - \pi) = -\pi/2$$

$$\therefore \text{Eq (1)} \Rightarrow$$

$$f(0) = -\frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{-1}{n^2 \pi} (1 - (-1)^n)$$

$$-\frac{\pi}{2} = -\frac{\pi}{4} - \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} (1 - (-1)^n)$$

$$\frac{\pi}{4} - \frac{\pi}{2} = -\frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} (1 - (-1)^n)$$

$$-\frac{\pi}{4} = -\frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} (1 - (-1)^n)$$

$$\frac{\pi^2}{4} = \sum_{n=1,3,5,\dots}^{\infty} \frac{2}{n^2}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

10) obtain the fourier series for the function

$$f(x) = \begin{cases} 1 + \frac{4x}{3} & \text{in } -3/2 < x \leq 0 \\ 1 - \frac{4x}{3} & \text{in } 0 \leq x < 3/2 \end{cases}$$

Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Solun:- To check given function even (or) odd

$$\text{Let } \phi(x) = 1 + \frac{4x}{3} \quad \psi(x) = 1 - \frac{4x}{3}$$

$$\text{Consider } \phi(-x) = 1 + \frac{4(-x)}{3} = 1 - \frac{4x}{3} = \psi(x)$$

$$\text{i.e. } \phi(-x) = \psi(x)$$

$\therefore f(x)$ is even function

$$\therefore \boxed{b_n = 0}$$

here $f(x)$ is defined in $(-\frac{3}{2}, \frac{3}{2})$
(-1, 1)

$$\therefore \boxed{l = 3/2}$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$= \frac{2}{3/2} \int_0^{3/2} \left(1 - \frac{4x}{3}\right) dx = \frac{4}{3} \left(x - \frac{4x^2}{6}\right)_0^{3/2}$$

$$= \frac{4}{3} \left[\left(\frac{3}{2} - \frac{4}{3 \times 2} \left(\frac{3}{2}\right)^2\right) - 0\right]$$

$$= \frac{4}{3} \left(\frac{3}{2} - \frac{3}{2}\right) = 0$$

$$\Rightarrow \boxed{a_0 = 0}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{3/2} \int_0^{3/2} \left(1 - \frac{4x}{3}\right) \cos\left(\frac{2n\pi x}{3}\right) dx$$

$$= \frac{4}{3} \left[\left(1 - \frac{4x}{3}\right) \left(\frac{\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)}\right) - \left(-\frac{4}{3}\right) \frac{\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2} \right]_0^{3/2}$$

$$= \frac{4}{3} \times \frac{-4}{3} \times \frac{9}{4n^2\pi^2} \left(\cos\left(\frac{2n\pi x}{3}\right)\right)_0^{3/2}$$

$$= \frac{-4}{n^2\pi^2} (\cos n\pi - 0)$$

$$= \frac{-4}{n^2\pi^2} ((-1)^n - 1)$$

$$\boxed{a_n = \frac{4}{n^2\pi^2} (1 - (-1)^n)}$$

The fourier series for $f(x)$ with period $2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$
$$= \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} (1 - (-1)^n) \cos\left(\frac{2n\pi x}{3}\right) \quad \text{--- ①}$$

is required fourier series

Deduction

put $x=0$ in eq ①

$$f(0) = \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} [1 - (-1)^n]$$

$$1 = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} (1 - (-1)^n)$$

$$\frac{\pi^2}{4} = \sum_{n=1,3,5,\dots}^{\infty} \frac{2}{n^2}$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

11) Find the Fourier series of the periodic function defined by $f(x) = 2x - x^2$ in the interval $0 < x < 3$.

Solution Given $f(x) = 2x - x^2$ & interval $(0, 3)$

Compare with $(0, 2l)$

$$\Rightarrow 2l = 3$$

$$\boxed{l = 3/2}$$

The Fourier series of $f(x)$ with period $2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \quad \text{--- (1)}$$

To check $f(x)$ is even (or) odd

we have $f(x) = 2x - x^2$ with $(0, 3)$

$$\text{Consider } f(3-x) = 2(3-x) - (3-x)^2$$

$$= 6 - 2x - (9 + x^2 - 6x) \neq f(x) \\ \neq -f(x)$$

$\therefore f(x)$ is neither even nor odd function.

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

$$= \frac{1}{3/2} \int_0^3 (2x - x^2) dx = \frac{2}{3} \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^3$$

$$= \frac{2}{3} \left(9 - \frac{27}{3} \right)$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{1}{l} \int_0^3 f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{3/2} \int_0^3 (2x - x^2) \cos\left(\frac{2n\pi x}{3}\right) dx$$

$$= \frac{2}{3} \left[(2x - x^2) \left(\frac{\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)} \right) - (2 - 2x) \times \left(\frac{-\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2} \right) \right. \right.$$

$$\left. + (-2) \left(\frac{-\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^3} \right) \right]_0^3$$

$$= \frac{2}{3} \times \frac{9}{4n^2\pi^2} \left[(2 - 2x) \cos\left(\frac{2n\pi x}{3}\right) \right]_0^3$$

$$= \frac{3}{2n^2\pi^2} [2 - 6) \cos(2n\pi) - 2 \cos(0)]$$

$$= \frac{3}{2n^2\pi^2} (-4 \times (1) - 2) = \frac{3 \times -6}{2n^2\pi^2}$$

$$\boxed{a_n = \frac{-9}{n^2\pi^2}}$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{3/2} \int_0^3 (2x - x^2) \sin\left(\frac{2n\pi x}{3}\right) dx$$

$$= \frac{2}{3} \left[(2x - x^2) \left(\frac{-\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)} \right) - (2 - 2x) \left(\frac{-\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2} \right) + (-2) \left(\frac{\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^3} \right) \right]_0^3$$

$$\begin{aligned}
 &= \frac{2}{3} \left[\frac{-3}{2n\pi} \left\{ (2x - x^2) \cos\left(\frac{2n\pi x}{3}\right) \right\}_0^3 - \frac{54}{8n^3\pi^3} \left(\cos\left(\frac{2n\pi x}{3}\right) \right)_0^3 \right] \\
 &= \frac{2}{3} \left[\frac{-3}{2n\pi} \left((6 - 9) \cos(2n\pi) - 0 \right) - \frac{54}{8n^3\pi^3} (\cos(2n\pi) - \cos 0) \right] \\
 &= \frac{2}{3} \left[\frac{-3}{2n\pi} (-3) \times 1 - \frac{54}{8n^3\pi^3} (1 - 1) \right] \\
 &= \frac{2}{3} \frac{+9}{2n\pi}
 \end{aligned}$$

$$b_n = \frac{3}{n\pi}$$

$\therefore$ Eq (1) $\Rightarrow$

$$f(x) = \sum_{n=1}^{\infty} \frac{-9}{n^2\pi^2} \cos\left(\frac{2n\pi x}{3}\right) + \sum_{n=1}^{\infty} \frac{3}{n\pi} \sin\left(\frac{2n\pi x}{3}\right)$$

is the required fourier series.

Half Range Fourier Series

Interval	Required Series	Series	Co-efficients
$(0, l)$	cosine series	$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$	$a_0 = \frac{2}{l} \int_0^l f(x) dx$ $a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$
	Sine Series	$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$	$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$
$(0, \pi)$	cosine series	$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$	$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$ $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$
	Sine Series	$\sum_{n=1}^{\infty} b_n \sin(nx)$	$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx$

1) Find the cosine series for $f(x) = (x-1)^2$ in $0 < x < 1$. Hence deduce $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Solution:- Given $f(x) = (x-1)^2$ in $(0, 1)$

Compare given interval with $(0, l)$

$$\Rightarrow \boxed{l=1}$$

$$\begin{aligned} a_0 &= \frac{2}{l} \int_0^l f(x) dx = \frac{2}{1} \int_0^1 (x-1)^2 dx = 2 \left[\frac{(x-1)^3}{3} \right]_0^1 \\ &= \frac{2}{3} [(1-1)^3 - (-1)^3] = \frac{2}{3} \end{aligned}$$

$$\boxed{a_0 = \frac{2}{3}}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{1} \int_0^1 (x-1)^2 \cos(n\pi x) dx$$

$$= 2 \left[(x-1)^2 \left(\frac{\sin(n\pi x)}{(n\pi)} \right) - 2(x-1) \left(\frac{\cos(n\pi x)}{(n\pi)^2} \right) + 2 \left(\frac{\sin(n\pi x)}{(n\pi)^3} \right) \right]_0^1$$

$$= 2 \times \frac{2}{n^2 \pi^2} \left[(x-1) \cos(n\pi x) \right]_0^1$$

$$= \frac{4}{n^2 \pi^2} ((1-1) \cos(n\pi) - (-1) \cos 0)$$

$$= \frac{4}{n^2 \pi^2} (0 + 1)$$

$$\therefore \boxed{a_n = \frac{4}{n^2 \pi^2}}$$

The fourier half range cosine series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2/3}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos(n\pi x)$$

$$= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos(n\pi x) \quad \text{--- (1)}$$

is required series

Deduction: put $x=0$ in eq (1)

$$f(0) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$1 - \frac{1}{3} = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\frac{2}{3} = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \quad \text{--- (2)}$$

Again put $x=1$ in eq (2).

$$f(1) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos(n\pi)$$

$$0 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} (-1)^n$$

$$-\frac{1}{3} = \frac{4}{\pi^2} \left[\frac{-1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots \right]$$

$$-\frac{\pi^2}{12} = -\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots$$

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \quad \text{--- (3)}$$

Adding (2) & (3).

we get

$$\frac{\pi^2}{6} + \frac{\pi^2}{12} = \frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \dots$$

$$\frac{3\pi^2}{12} = 2 \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$$

$$\Rightarrow \boxed{\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots}$$

2) Find the cosine half range series for
 $f(x) = x(l-x)$; $0 \leq x \leq l$.

Solution:- Here $f(x)$ defined in $(0, l)$

$$\begin{aligned} a_0 &= \frac{2}{l} \int_0^l f(x) dx = \frac{2}{l} \int_0^l x(l-x) dx \\ &= \frac{2}{l} \int_0^l (xl - x^2) dx \\ &= \frac{2}{l} \left[l \frac{x^2}{2} - \frac{x^3}{3} \right]_0^l \\ &= \frac{2}{l} \left(l \frac{l^2}{2} - \frac{l^3}{3} \right) = \frac{2}{l} l^3 \left(\frac{1}{2} - \frac{1}{3} \right) \\ &= 2 l^2 \frac{1}{6} \Rightarrow \boxed{a_0 = \frac{l^2}{3}} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{2}{l} \int_0^l x(l-x) \cos\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{2}{l} \int_0^l (lx - x^2) \cos\left(\frac{n\pi x}{l}\right) dx \\ &= \frac{2}{l} \left[\cancel{lx - x^2} \left(\frac{\sin\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)} \right) - (l-2x) \left(\frac{-\cos\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^2} \right) - 2 \left(\frac{-\sin\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^3} \right) \right]_0^l \end{aligned}$$

$$= \frac{2}{l} \left[\frac{l^2}{n^2 \pi^2} \left\{ (l-2x) \cos\left(\frac{n\pi x}{l}\right) \right\}_0^l \right]$$

$$= \frac{2 l^2}{l n^2 \pi^2} \left[(l-2l) \cos(n\pi) - l \cos 0 \right]$$

$$= \frac{2l}{n^2 \pi^2} (-l(-1)^n - l)$$

$$\boxed{a_n = -\frac{2l^2}{n^2 \pi^2} [1 + (-1)^n]}$$

The fourier half range cosine series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$= \frac{l^2/3}{2} + \sum_{n=1}^{\infty} \frac{-2l^2}{n^2 \pi^2} [1 + (-1)^n] \cos\left(\frac{n\pi x}{l}\right)$$

$$f(x) = \frac{l^2}{6} - \frac{2l^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 + (-1)^n] \cos\left(\frac{n\pi x}{l}\right)$$

3) If $f(x) = \begin{cases} x & \text{in } 0 < x < \pi/2 \\ \pi - x & \text{in } \pi/2 < x < \pi \end{cases}$, find the half range fourier sine series of $f(x)$.

Solution:-

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx \\
 &= \frac{2}{\pi} \left[\int_0^{\pi/2} f(x) \sin(nx) dx + \int_{\pi/2}^{\pi} f(x) \sin(nx) dx \right] \\
 &= \frac{2}{\pi} \left[\int_0^{\pi/2} x \sin(nx) dx + \int_{\pi/2}^{\pi} (\pi - x) \sin(nx) dx \right] \\
 &= \frac{2}{\pi} \left\{ \left[x \left(\frac{-\cos(nx)}{n} \right) - \left(\frac{-\sin(nx)}{n^2} \right) \right]_0^{\pi/2} + \left[(\pi - x) \left(\frac{-\cos(nx)}{n} \right) - (-1) \left(\frac{-\sin(nx)}{n^2} \right) \right]_{\pi/2}^{\pi} \right\} \\
 &= \frac{2}{\pi} \left[-\frac{1}{n} (x \cos nx) \Big|_0^{\pi/2} + \frac{1}{n^2} (\sin nx) \Big|_0^{\pi/2} - \frac{1}{n} ((\pi - x) \cos nx) \Big|_{\pi/2}^{\pi} - \frac{1}{n^2} (\sin nx) \Big|_{\pi/2}^{\pi} \right] \\
 &= \frac{2}{\pi} \left[-\frac{1}{n} \left(\frac{\pi}{2} \cos\left(\frac{n\pi}{2}\right) - 0 \right) + \frac{1}{n^2} \left(\sin\left(\frac{n\pi}{2}\right) - 0 \right) - \frac{1}{n} \left((\pi - \pi) \cos n\pi - \frac{\pi}{2} \cos\left(\frac{n\pi}{2}\right) \right) - \frac{1}{n^2} \left(0 - \sin\left(\frac{n\pi}{2}\right) \right) \right] \\
 &= \frac{2}{\pi} \left[\frac{-\pi}{2n} \cos\left(\frac{n\pi}{2}\right) + \frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right) + \frac{\pi}{2n} \cos\left(\frac{n\pi}{2}\right) + \frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right) \right] \\
 &= \frac{2}{\pi} \left[\frac{2}{n^2} \sin\left(\frac{n\pi}{2}\right) \right]
 \end{aligned}$$

$$b_n = \frac{4}{\pi n^2} \sin\left(\frac{n\pi}{2}\right)$$

The fourier half range sine series is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx) \\ = \sum_{n=1}^{\infty} \frac{4}{\pi n^2} \sin\left(\frac{n\pi}{2}\right) \sin(nx)$$

4) obtain the sine half range series of

$$f(x) = \begin{cases} \frac{1}{4} - x & \text{in } 0 < x < \frac{1}{2} \\ x - \frac{3}{4} & \text{in } \frac{1}{2} < x < 1 \end{cases}$$

Soln:- Here $f(x)$ is defined in $(0,1)$.

Comparing with $(0,l) \Rightarrow \boxed{l=1}$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \\ = \frac{2}{1} \left[\int_0^{1/2} f(x) \sin(n\pi x) dx + \int_{1/2}^1 f(x) \sin(n\pi x) dx \right] \\ = 2 \left[\int_0^{1/2} \left(\frac{1}{4} - x\right) \sin(n\pi x) dx + \int_{1/2}^1 \left(x - \frac{3}{4}\right) \sin(n\pi x) dx \right] \\ = 2 \left\{ \left[\left(\frac{1}{4} - x\right) \left(-\frac{\cos(n\pi x)}{n\pi}\right) - (-1) \left(-\frac{\sin(n\pi x)}{(n\pi)^2}\right) \right]_0^{1/2} \right. \\ \left. + \left[\left(x - \frac{3}{4}\right) \left(-\frac{\cos(n\pi x)}{n\pi}\right) - 1 \times \left(-\frac{\sin(n\pi x)}{(n\pi)^2}\right) \right]_{1/2}^1 \right\}$$

$$= 2 \left[-\frac{1}{n\pi} \left(\left(\frac{1}{4} - x \right) \cos n\pi \right) \Big|_0^{1/2} - \frac{1}{n^2\pi^2} \left\{ \sin(n\pi x) \right\} \Big|_0^{1/2} - \frac{1}{n\pi} \left\{ \left(x - \frac{3}{4} \right) \cos(n\pi x) \right\} \Big|_{1/2}^{1/2} + \frac{1}{n^2\pi^2} \left\{ \sin(n\pi x) \right\} \Big|_{1/2}^{1/2} \right]$$

$$= 2 \left[\left(-\frac{1}{n\pi} \right) \left\{ \frac{1}{4} \cos\left(\frac{n\pi}{2}\right) - \frac{1}{4} \cos(0) \right\} - \frac{1}{n^2\pi^2} \left(\sin\left(\frac{n\pi}{2}\right) - 0 \right) - \frac{1}{n\pi} \left\{ \left(1 - \frac{3}{4} \right) \cos(n\pi) - \left(\frac{1}{2} - \frac{3}{4} \right) \cos\left(\frac{n\pi}{2}\right) \right\} + \frac{1}{n^2\pi^2} \left\{ \sin(n\pi) - \sin\left(\frac{n\pi}{2}\right) \right\} \right]$$

$$= 2 \left[\frac{1}{4n\pi} \left(\cos\frac{n\pi}{2} + 1 \right) - \frac{1}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) - \frac{1}{n\pi} \left\{ +\frac{1}{4} \cos(n\pi) + \frac{1}{4} \cos\left(\frac{n\pi}{2}\right) \right\} + \frac{1}{n^2\pi^2} \left(-\sin\left(\frac{n\pi}{2}\right) \right) \right]$$

$$= 2 \left[\frac{1}{4n\pi} \left(\cos\left(\frac{n\pi}{2}\right) + 1 \right) - \frac{1}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) + \frac{1}{4n\pi} \left((-1)^n - \cos\left(\frac{n\pi}{2}\right) \right) \right]$$

$$= 2 \left[\frac{1}{4n\pi} \left(\cancel{\cos\left(\frac{n\pi}{2}\right)} + 1 + (-1)^n - \cancel{\cos\left(\frac{n\pi}{2}\right)} \right) - \frac{2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \right]$$

$$= 2 \left[\frac{1}{4n\pi} \left(1 - (-1)^n \right) - \frac{2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \right]$$

$$b_n = \frac{1}{2n\pi} \left(1 - (-1)^n \right) - \frac{4}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

The fourier half range sine series is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \sum_{n=1}^{\infty} \left[\frac{1}{2n\pi} (1 - (-1)^n) - \frac{4}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \right] \cdot \sin(n\pi x)$$

VTUSYNC.IN

Practical Harmonic Analysis

Harmonic Analysis is the process of finding the constant term & the first few cosine & sine terms numerically

The Fourier Series of $f(x)$ in

$$[0, 2\pi), \text{ or } (0, 2\pi]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0 = \frac{2}{N} \sum y ; a_n = \frac{2}{N} \sum y \cos nx ;$$

$$b_n = \frac{2}{N} \sum y \sin nx$$

$$[0, 2l) \text{ or } (0, 2l]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{2}{N} \sum y ;$$

$$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$$

$$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$$

$N = N_0$ of values given in table

$$\theta = \frac{\pi x}{l}$$

1) Determine the constant term & the first cosine & sine terms of the fourier series expansion of y from the following table.

(or)
obtain the fourier series neglecting terms higher than first harmonic.

x°	0	60°	120°	180°	240°	300°
y	7.9	7.2	3.6	0.5	0.9	6.8

Solution:-

Here the interval of x is 0° to 360°

i.e. $0 \leq x < 2\pi$ $[0, 2\pi)$

The series containing the first harmonic is

$$y = f(x) = \frac{a_0}{2} + a_1 \cos nx + b_1 \sin nx \quad \text{--- (1)}$$

x	y	$\cos x$	$y \cos x$	$\sin x$	$y \sin x$
0	7.9	1	7.9	0	0
60°	7.2	0.5	3.6	0.866	6.2352
120°	3.6	-0.5	-1.8	0.866	3.1176
180°	0.5	-1	-0.5	0	0
240°	0.9	-0.5	-0.45	-0.866	-0.7794
300°	6.8	0.5	3.4	-0.866	-5.8888
Totals	26.9		12.15		2.6846

$$N=6$$

$$a_0 = \frac{2}{N} \sum y = \frac{2}{6} (26.9) = 8.9667$$

$$a_1 = \frac{2}{N} \sum y \cos x = \frac{2}{6} (12.15) = 4.05$$

$$b_1 = \frac{2}{N} \sum y \sin x = \frac{2}{6} (2.6846) = \frac{1}{3} (2.6846) = 0.8949$$

$\therefore \text{Eq. (1)} \Rightarrow$

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x$$

$$= \frac{8.9667}{2} + 4.05 \cos x + 0.8949 \sin x$$

$$f(x) = 4.48335 + (4.05) \cos x + (0.8949) \sin x$$

2) Compute the constant term & first two harmonics of the fourier series of $f(x)$ given the following table

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Soln:-

Here $f(x)=y$ defined in the interval $0 \leq x \leq 2\pi$

$f(0) = f(2\pi) = 1.0$ so we omit the last value of $f(x)$.

x	y	$\cos x$	$\cos 2x$	$\sin x$	$\sin 2x$	$y \cos x$	$y \cos 2x$	$y \sin x$	$y \sin 2x$
0	1.0	1	1	0	0	1	1	0	0
$\pi/3$	1.4	0.5	-0.5	0.866	0.866	0.7	-0.7	1.2124	1.2124
$2\pi/3$	1.9	-0.5	-0.5	0.866	-0.866	-0.95	-0.95	1.6454	-1.6454
π	1.7	-1	1	0	0	-1.7	1.7	0	0
$4\pi/3$	1.5	-0.5	-0.5	-0.866	0.866	-0.75	-0.75	-1.299	1.299
$5\pi/3$	1.2	0.5	-0.5	-0.866	-0.866	0.6	-0.6	-1.0392	-1.0392
Totals	8.7					-1.1	-0.3	0.5196	-0.132

Here $N=6$

$$a_0 = \frac{2}{N} \sum y = \frac{2}{6} (8.7) = 2.9$$

$$a_1 = \frac{2}{N} \sum y \cos x = \frac{2}{6} (-1.1) = -0.367$$

$$a_2 = \frac{2}{N} \sum y \cos 2x = \frac{2}{6} (-0.3) = -0.1$$

$$b_1 = \frac{2}{N} \sum y \sin x = \frac{2}{6} (0.5196) = 0.1732$$

$$b_2 = \frac{2}{N} \sum y \sin 2x = \frac{2}{6} (-0.1732) = -0.0577$$

The first two harmonics are

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x)$$

$$\# \frac{2.9}{2} + (-0.367 \cos x + 0.1732 \sin x) +$$

$$= \frac{2.9}{2} + [-0.367 \cos x + 0.1732 \sin x] + [-0.1 \cos 2x + (-0.0577) \sin 2x]$$

$$= 1.45 + [-0.367 \cos x + 0.1732 \sin x] + [-0.1 \cos 2x - 0.0577 \sin 2x]$$

$$\therefore \text{Constant term } \left(\frac{a_0}{2}\right) = 1.45$$

$$\text{First harmonics} = [-0.367 \cos x + 0.1732 \sin x]$$

$$\text{Second harmonics} = [-0.1 \cos 2x - 0.0577 \sin 2x]$$

3) obtain the constant term & the co-efficients of the first cosine & sine terms in the fourier expansion of y from the table

x	0	1	2	3	4	5
y	9	18	24	28	26	20

Solun: Here $N=6$

and the interval should be $0 \leq x < 6$

length of the interval is $6-0=6$.

Comparing with $2l$

$$\Rightarrow 2l=6 \quad \Rightarrow \boxed{l=3}$$

the fourier series of period $2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

since $l=3$, the series containing the first harmonics is

$$f(x) = \frac{a_0}{2} + a_1 \cos\left(\frac{\pi x}{3}\right) + b_1 \sin\left(\frac{\pi x}{3}\right)$$

$$\text{Put } \frac{\pi x}{3} = \theta$$

$$\therefore f(x) = \frac{a_0}{2} + a_1 \cos \theta + b_1 \sin \theta \quad \text{--- (1)}$$

x	$\theta = \frac{\pi x}{3}$	y	$\cos \theta$	$y \cos \theta$	$\sin \theta$	$y \sin \theta$
0	0	9	1	9	0	0
1	60°	18	0.5	9	0.866	15.588
2	120°	24	-0.5	-12	0.866	20.784
3	180°	28	-1	-28	0	0
4	240°	26	-0.5	-13	-0.866	-22.516
5	300°	20	0.5	10	-0.866	-17.32
Total		125		-25		-3.464

Here $N=6$

$$a_0 = \frac{2}{N} \sum y = \frac{2}{6} (125) = 41.67$$

$$a_1 = \frac{2}{N} \sum y \cos \theta = \frac{2}{6} (-25) = -8.333$$

$$a_2 = \frac{2}{N} \sum y \sin \theta = \frac{2}{6} (-3.464) = -1.155$$

$\therefore$ Eq (1) $\Rightarrow$

$$f(x) = \frac{41.67}{2} + [(-8.333) \cos x + (-1.155) \sin x]$$

$$f(x) = 20.835 - 8.333(\cos x) - 1.155(\sin x)$$

$$\therefore \text{Constant term} = \frac{a_0}{2} = 20.835$$

$$\text{Co-efficient of the first cosine term} = a_1 = -8.333$$

$$\text{Co-efficient of the first sine term} = b_1 = -1.155$$

4) The following table gives the variations of periodic current over a period.

t(sec)	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A(amps)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Show that there is a direct current part of 0.75 amp.
 & obtain the amplitude of the first harmonic.

Soln:- We observe that

$$A(0) = A(T) = 1.98.$$

Hence neglect the last value.

convert $A=f(t)$ to the period 2π by

putting

$$\theta = \frac{\pi t}{l}$$

$$\text{here } \boxed{l = \frac{T}{2}}$$

$$= \frac{\pi t}{T/2} = \frac{2\pi t}{T}$$

$$\text{ie } \boxed{\theta = \frac{2\pi t}{T}}$$

The fourier series upto first harmonic is

$$A = \frac{a_0}{2} + (a_1 \cos \theta + b_1 \sin \theta) \quad \text{--- (1)}$$

t	$\theta = \frac{2\pi T}{t}$	A	$\cos \theta$	$\sin \theta$	$A \cos \theta$	$A \sin \theta$
0	0°	1.98	1	0	1.98	0
$T/6$	60°	1.30	0.5	0.866	0.65	1.258
$T/3$	120°	1.05	-0.5	0.866	-0.525	0.9093
$T/2$	180°	1.30	-1	0	-1.30	0
$2T/3$	240°	-0.88	-0.5	-0.866	-0.44	0.7621
$5T/6$	300°	-0.25	0.5	-0.866	-0.125	0.2165
Total		4.5			1.12	3.0137

Here $N=6$

$$a_0 = \frac{2}{N} \sum A = \frac{2}{6} (4.5) = 1.5$$

$$a_1 = \frac{2}{N} \sum A \cos \theta = \frac{2}{6} (1.12) = 0.3733$$

$$a_2 = \frac{2}{N} \sum A \sin \theta = \frac{2}{6} (3.0137) = 1.0046$$

$$\therefore \text{eqn } \Rightarrow A = \frac{1.5}{2} + \left[(0.3733) \cos x + (1.0046) \sin x \right]$$

$$= 0.75 + \left[(0.3733) \cos x + (1.0046) \sin x \right]$$

The direct current part of the variable is the constant term $\left(\frac{a_0}{2} \right) = 0.75$

$$\text{Amplitude of the first harmonics} = \sqrt{a_1^2 + b_1^2} = \sqrt{(0.3733)^2 + (1.0046)^2}$$

$$= 1.0717$$